

Nick Eisenberg's Meeting Notes

Le Chen, Nick Eisenberg, Zhijian Wu

October 2019

1 Introduction

This is a document to keep track of meetings between Nick Eisenberg, Dr. Zhijian Wu and Dr. Le Chen. We can use the following environments to pose questions.

Nick's Question 1.1 (Spectral measure) Here is a question by Nick related to spectral measure.

Le's solution. Le's sketch of proof is here. ☐

Zhijian's suggestion. Zhijian's suggestions go here. ☐

Nick's solution. Here is the proof by Nick. ☐

Le's Question 1.1 Here is a question by Le.

Nick's solution. Here is the proof by Nick. ☐

Zhijian's Question 1.1 (White noise) Here is a question by Zhijian.

Nick's solution. Here is the proof by Nick. ☐

2 Meeting Oct. 8

3 Meeting Oct. 22

Nick's Question 3.1 (delta)

Nick's Question 3.2

4 Meeting Oct. 29

Le's Question 4.1 We are interested in stochastic heat equation:

$$\begin{cases} \left(\frac{\partial}{\partial t} - \frac{1}{2} \Delta \right) u(t, x) = \rho(u(t, x)) \dot{W}(t, x), & t > 0, x \in \mathbb{R}^d, \\ u(0, \cdot) = \mu, \end{cases} \quad (1)$$

where ρ is Lipschitz continuous and $\dot{W}$ is some noise that is white in time and colored

in space. The SHE (1), sometimes with a nonlinear drift term $b(u(t, x))$ on the right-hand side, describes systems in physics and mathematical biology, where there is a strong need to know the ergodic behavior of such systems. The first step in understanding the ergodicity of these systems is, of course, to show the existence of an invariant measure for the corresponding dynamical system. There is a rich literature for the study of invariant measure in case of bounded spatial domain; see, e.g., [6, 10, 11, 12, 21, 25, 29]. However, the case of unbounded domain has been much less studied; see [2, 22, 30, 39].

In this preliminary project, we will focus on the case when $b \equiv 0$ and the spatial domain is unbounded, namely, $\mathbb{R}^d$. The case with nonvanishing drift term b will be studied in the future upon completion of this project. For the case of unbounded domain, one has to include some *weight functions*. Let ρ be a positive bounded function in $L^1(\mathbb{R}^d)$, associated with which is the weighted $L^2_\rho(\mathbb{R}^d)$ space with the norm $\|\phi\|_\rho^2 = \int_{\mathbb{R}^d} \phi^2(x) \rho(x) dx$. Following [39], the problem reduces to prove that for any fixed $T > 0$, $\|u\|_{2,T,\rho}^2 := \sup_{t \in [0,T]} \mathbb{E}(\|u(t, \cdot)\|_\rho^2) < \infty$. Notice that $\|u\|_{2,T,\rho}^2 = \sup_{t \in [0,T]} \int_{\mathbb{R}^d} \|u(t, x)\|_2^2 \rho(x) dx$. Now we can apply our sharp moment formula/bounds:

$$\|u(t, x)\|_p \leq C [1 + (|\mu| * G(t, \cdot)(x))] H_p^{1/2}(t) \quad (2)$$

from [15]. Tessitore and Zabczyk [39] studied this problem under strong conditions on: (1) the correlation functions Λ , which exclude the important Riesz kernel case, (2) the weight functions (polynomial decay or exponential decay), and (3) the initial data (in some Weighted $L^2_\rho(\mathbb{R}^d)$ space). Accordingly, with help of our sharp moment formula/bounds, we will be able to relax their conditions to allow (1) more general correlation function Λ (including the Riesz kernel); (2) smaller admissible weight functions (e.g., $\rho(x) = e^{-c_T|x|^2}$ for some sufficiently large constant $c_T > 0$, note that the smaller is ρ , the larger is the space $L^2_\rho(\mathbb{R}^d)$); (3) rough initial data. In particular, our extension will include the PAM with, e.g., delta initial data.

5 Meeting on Nov. 5th

Le's Question 5.1 Regarding the two approaches — the random field approach (ours) and semigroup approach, you may read paper by Dalang and Quer-Sardanyons [19] for the equivalence of the two approaches. You may also take a brief look at Da Prato's book [20], the counterpart of the Walsh notes for random field approach, for the semigroup approach.

Nick's Question 5.1 (spatial variable of the solution/ what type of object is the solution) In [39] we write the solution of (1.1) as (2.3). More specifically we write the solution as

$$X^x(t) \quad \text{for } x \in L^2_\rho \text{ and } t > 0$$

but we lose the spatial component of X from (1.1) namely we no longer write X as

$$X^x(t, \xi) \quad \text{for } x \in L^2_\rho, \xi \in \mathbb{R}^d \text{ and } t > 0$$

however by Theorem 2.3 of [39] we still know that the solution to (1.1) is in L^2_ρ for each $\omega \in \Omega$. Namely for each $x \in L^2_\rho$, $t > 0$, and $\omega \in \Omega$ then $X^x(t)(\omega) \in L^2_\rho$. So does this mean that the solution in fact does have a spatial element but we don't write it since we lose this information about it? What I mean by this is that according to the Theorem 2.3, our solution $X^x(\cdot) \in \mathcal{C}_{\mathcal{P}}(0, T, L^2(\Omega, L^2_\rho))$ which is the set of all predictable processes

$Y : \Omega \times [0, T] \rightarrow L_\rho^2$ with the property that $t \mapsto E(|Y(t)|_\rho^2)$ is continuous so you can see here that we are integrating out the spatial variable when taking the $|\cdot|_\rho^2$ norm.

Nick's solution. After careful reading of the Da Prato text [21] and the Tessitore paper [39]. I am fairly confident that what I am saying above is correct. $\square$

Le's comment. You are right. From here, you see that Da Prato's approach is really an infinite-dimensional SDE. In the finite dimensional case, you simply replace the $|\cdot|_\rho$ norm by Euclidean norm. $\square$

Nick's Question 5.2 (Law of the solution) Now how are we defining the Law of the solution? Are we saying that

$$\mathcal{L}(X^x(t))(A) = \mathbf{P}\{\omega \in \Omega : X^x(t)(\omega) \in A \text{ for } A \subset L_\rho^2\} \quad (3)$$

so we are fixing t and x and the law of X is a probability measure on L_ρ^2 ? So if this was the case then by [21] on page 89 equation (6.1.2) we see if $\frac{1}{T} \int_0^T \mathcal{L}(X^x(t))dt$ is tight then that there exists an invariant measure for L_ρ^2 . (Also I have the details worked out which proves the statement about equation (6.1.2))

However there is a minor discrepancy here when comparing the Law defined the way I just had defined it above and how [21] defines it. Again on page 89 of [21] they define the law to be a measure on $\mathcal{B}(H)$ but does not define what $\mathcal{B}(H)$ is. Im guessing that it means the space of bounded linear functionals on some hilbert space H . The Law I defined above would be defined on $H = L_\rho^2$ not $\mathcal{B}(H) = \mathcal{B}(L_\rho^2)$. We are looking for a measure on L_ρ^2 and so I believe the way I had defined it above is how it is supposed to be defined in the Tessitore paper [39].

Nick's Solution. The law of X is what I defined it to be in equation (5.1) and this can be seen from (3.2) on page 276 of the Tessitore paper [39]. $\square$

Le's Comment. In equation (5.1), you cannot take arbitrary subsets A from L_ρ^2 . It has to have certain measurability property, like Borel sets in the finite dimensional case. That's why in the book [21], they use $\mathcal{B}(H)$. Please double check the meaning of this notation in their book. $\square$

Nick's Question 5.3 The integral in (2.3) shown below

$$X^x(t) = S(t)x + \int_0^t S(t-s)M(B(X^x(s)))dW_s \quad (4)$$

is used in the proof however in (5.2) they turn $\mathscr{W}$ into W for the integrator. They claim that this follows from [37] but I could not find an explanation for it in there. In addition the M function in this paper [39] is not the same as the M function in [37].

Le's comment. Study Da Prato's red book [20] and [19]. $\square$

Nick's Question 5.4 (Proof of Theorem 3.1)

In **Step 1** of the proof, Tessitore defines an operator

$$F_\alpha f = \int_0^1 (1-s)^{\alpha-1} S(1-s)f(s)ds \quad (5)$$

and says that this operator is compact from $L^q[(0,1), L_\rho^2]$ to L_ρ^2 . First, what is $L^q[(0,1), L_\rho^2]$? Second how is $F_\alpha f \in L_\rho^2$? It seems that the integrand in a function of s and we are integrating out the s variable so there are no variables left which would just make $F_\alpha f \in \mathbb{R}$. Also, does $S(1-s)f(s) = (G(1-s, \cdot) * f(\cdot))(s)$?

Proof. I believe the answer to why $F_\alpha f$ is in L_ρ^2 is essentially the same reason as my solution for my question (5.5), ie. it comes from Da Pratos construction of the stochastic integral in [37]. Also I believe that $L^q[(0,1), L_\rho^2]$ is the space of predictable processes, $f(s)$ such that $f : (0,1) \times \Omega \rightarrow L_\rho^2$ and $\int_0^1 |f(s)|_\rho^q ds < \infty$. This can be verified by comparing the $|h|_{L^q(0,1,L_\rho^2)}^q$ in the definition of $\mathcal{K}(\theta)$ with integral on the right hand side of the first inequality after the mention of Chebyshev's inequality. $\square$

Nick's Question 5.5 (Proof of Theorem 3.1) In **Step 2** the $Y^x(s)$ that is defined is apparently in L_ρ^2 since in this step we use $|Y^x(s)|_\rho^2$. However from the definition of Y in the beginning of step 2, I don't see why $Y^x(s) \in L_\rho^2$. Also I am having trouble in filling in the details for the inequality

$$E \int_0^1 |Y^x(s)|_\rho^q ds \leq k \int_0^1 \left(E \int_0^s (s-\varsigma)^{-2\alpha} \|S(s-\varsigma)M(B(X^x(\varsigma)))\|_{L_2(L^2, L_\rho^2)}^2 d\varsigma \right)^{q/2} ds \quad (6)$$

Nick's Solution. As for the reason why $Y \in L_\rho^2$, I believe this comes from Da Prato's construction of the stochastic integral $\int_0^t \psi(x) dW_x$, see and page 57 of [21]. In genera we can define the stochastic integral $\int_0^t \psi(x) dW_x$ when ψ is a Hilbert schmidt operator from a Hilbert space U to another Hilbert space H . In this paper, the integrand of $Y^x(s)$ is playing the role of ψ but here it is a hilbert schmidt operator from $L^2 \rightarrow L_\rho^2$ and this is why our stochastic integral Y takes values in L_ρ^2 .

The inequality follows from (5.2.13) from [21] and also page 192 of [37]. Ie. we can use (5.2.13) to show that

$$E \int_0^1 |Y^x(s)|_\rho^q ds \leq k \int_0^1 E \left(\int_0^s (s-\varsigma)^{-2\alpha} \|S(s-\varsigma)M(B(X^x(\varsigma)))\|_{L_2(L^2, L_\rho^2)}^2 d\varsigma \right)^{q/2} ds \quad (7)$$

$$\leq kC^q \int_0^1 E \left(\int_0^s (s-\varsigma)^{-(2\alpha+\sigma)} |B(X^x(\varsigma))|_\rho^2 d\varsigma \right)^{q/2} ds \quad (8)$$

where σ is defined in the in the paper and comes from Proposition 2.4 on the previous page. Also there is a small type in (5.4) which is the inequality how it is written in the paper. The E needs to be outside of the $(\cdot)^{q/2}$ and there is also a missing power of 2 on the $\|\cdot\|$ term in the integrand in (5.4). I fixed both of these in (5.5). $\square$

Nick's Question 5.6 On page 279 in the proof of Theorem 3.3, I am unable to see

why

$$l_0^2 \int_0^\infty \int_{\mathbb{R}^d} \left((G(s, \cdot) * |\widehat{\gamma^{1/2}}|)(z) \right)^2 dz ds = \int_0^\infty \int_{\mathbb{R}^d} G(2s, z) \tilde{\Gamma}(z) dz ds \quad (9)$$

Nick's Question 5.7 What is the β_s^i integrator and how do we obtain the moment estimate for Z_1^2

Nick's Question 5.8 How do we get the last equality on the top of page 280. This is similar to my question 5.6. as well as the moment estimate for Z_2^2 ?

Le's Question 5.2 Nick, I saw your questions. I will go over them this evening or tomorrow morning. Talk to you tomorrow then.

6 Meeting on Nov. 12th

Le's Question 6.1 Keep reading the paper by Tessitore and Zabczyk [39] and try to figure out more details in the paper.

Le's Question 6.2 See Remark 3.6 of [39]. Let's at least prove the same result, i.e., Theorem 3.3 of [39] under condition that

$$\int_{\mathbb{R}^d} \Gamma(y) |y|^{2-d} dy < \infty, \quad (10)$$

where $\Gamma(\cdot)$ is the correlation function as in [39]. Check my paper with Kunwoo Kim [16]. Lemma 2.5 of [16] gives the condition when $H(t)$ is bounded, which translates to

$$d \geq 3 \quad \text{and} \quad \int_{\mathbb{R}^d} \frac{f(x)}{|x|^{d-2}} dx < \infty; \quad (11)$$

see Eq. (1.12) of [16]. Here f is the spatial correlation function, which is Γ in [39]. Then instead of Hypothesis 2.1 (i) of [39], we only need L_ρ (the Lipschitz constant of ρ in (1)) is small enough.

Here is my conjecture and please try to prove the following statement:

Nick's Question 6.1 Should the integral in (6.10) be written as

$$\int_{\mathbb{R}^d} \Gamma(dy) |y|^{2-d}$$

as it is in chapter 2 of the mini course? And then f in (6.2) is the spectral density of Γ ? Then from using the relation that $\Gamma = \mathcal{F}\mu$ where $\mu(dx) = f(x)dx$ we see that

$$\int_{\mathbb{R}^d} \Gamma(dy) |y|^{2-d} = k \int_{\mathbb{R}^d} \frac{f(x)}{|x|^2} dx$$

where k is the constant from example 4.1 from page 55 of the mini course.

Conjecture Under condition (11), let L_ρ be the Lipschitz constant of ρ (see (1)), if L_ρ is sufficiently small, for any rough initial data and weight functions $w(\cdot)$ (more general than those in [39]), we have that

$$\sup_{t \geq 0} \mathbb{E} (|u^\mu(t)|_w^2) < \infty,$$

where μ denotes the rough initial data (not necessarily to be nonnegative).

Here are a list contributions:

- Initial data can be more general.
- I believe condition (11) small is more general than Hypothesis 2.1 (i) of [39]. See Question 6.3 below. Nevertheless, we have one additional condition, namely, L_ρ should be small enough.

Le's Question 6.3 Figure out the difference between condition (11) and Hypothesis 2.1 (i) of [39]. Why (11) is more general? Can you give several examples?

Nick's Question 6.2 Im a little confused, is our goal to drop hypothesis 2.1 (i) and make up for it by putting a restriction on L_ρ but in doing so we can use more general initial data? Hypothesis 2.1 (i) is used in the assumption of proposition 2.4 which is used to prove Theorem 3.1 which showed the existence of an invariant measure so if we drop hypothesis 2.1 (i) then we would have to reprove this result but I dont think this result was used in Theorem 3.3.

How about the condition (2.1) on the weight functions. Do we still want to assume that they satisfy the property that $G(t, \cdot) * \rho < C(T)\rho$ for every $t \in [0, T]$? Also, condition (2.1) was only used for proposition 2.4 which was only used to prove Theorem 3.1 so if out only goal is to focus on Theorem 3.3 the nI believe we can drop assumption (2.1) on the weight functions.

Le's comment . The question for condition (2.1) is slightly trickier. I wish we could do better than $\rho(x) = e^{-r|x|}$ for $r > 0$. For example, it would be good if we can cover $\rho(x) = e^{-|x|^\alpha}$ with $\alpha \in (1, 2)$. For such weight function, I don't believe condition (2.1) is true anymore. This will change slightly the statements in Proposition 2.1. $\square$

Nick's Question 6.3

Where in Theorem 3.3 is hypothesis 2.1 (i) used?

(*nick's proof*). I believe it is only indirectly used to show that the stochastic integrals are defined in the proof. Proposition 2.4 says that if we assume hypothesis 2.1 (i) then for $\phi \in L^2_\rho$ we have $S(t)M(\phi)$ is a Hilbert Schmidt operator from $L^2 \rightarrow L^2_\rho$ which is needed to ensure all the $\{X_n\}_{n \in \mathbb{N}}$ defined in the proof are well defined stochastic integrals. $\square$

Le's comment. This is no longer relevant. We can establish Theorem 3.3 using our language and under possibly weaker conditions on the spectral measure. $\square$

Nick's Question 6.4

I still think there are several inconsistencies with this paper that I can't justify.

1. First, this paper claims that the solution (2.3) on page 275 follows from [37] but in that paper the M is defined differently and I have found no justification in any other text as to why the M is defined differently in this paper.
2. This paper states that $\mathscr{W}$ is a Wiener Processes on L^2 with covariance operator I which I am assuming is the identity operator which according to De Prato means that $\mathscr{W}$ is a cylindrical Wiener processes. But this paper never states the range of $\mathscr{W}$ but in order for the stochastic integral in (2.3) to makes sense, $\mathscr{W}$ needs to be L^2 valued since our integrand in (2.3) is a hilbert schmidt operator from L^2 to L^2_ρ .
3. I havent quite been able to verify the representation of $Z_1(t, \xi)$ on page 279

$$Z_1(t, \xi) = \sum_i \int_0^t \left(\int_{\mathbb{R}^d} G(s, \xi - \eta) b(\eta, 0) (\widehat{\gamma^{1/2} * e_i})(\eta) d\eta \right) d\beta_s^i \quad (12)$$

Here's what I have been able to verify. First, sine the covariane operator is I , we use (4.2) page 82 of [20] to write

$$W(s) = \sum_i \langle W(s), e_i \rangle e_i \quad (13)$$

where $\{e_i\}$ is the orthonormal basis (not sure the space it is a basis for?). Then we use the definition of Z_1 given in the beginning of the proof as $Z_n = X_n - X_{n-1}$. The

proof says we let $X_0 = 0$. Then we have the following

$$Z_1(t, \xi) = X_1(t, \xi) = 1 + \int_0^t S(t-s)M(B(0))dW_s \quad (14)$$

$$= 1 + \sum_i \int_0^t S(t-s)M(B(0))d\langle W(s), e_i \rangle \quad (15)$$

$$= 1 + \sum_i \int_0^t S(t-s)M(B(0))e_i d\langle W(s), e_i \rangle \quad (16)$$

$$= 1 + \sum_i \int_0^t S(t-s)M(B(0))e_i d\beta_s^i \quad (17)$$

$$= 1 + \sum_i \int_0^t S(t-s)B(0)(\eta)(\widehat{\gamma^{1/2} * e_i})(\eta) d\beta_s^i \quad (18)$$

$$= 1 + \sum_i \int_0^t \left(\int_{\mathbb{R}^d} G(t-s, \xi - \eta) b(\eta, 0)(\widehat{\gamma^{1/2} * e_i})(\eta) d\eta \right) d\beta_s^i \quad (19)$$

However (6.10) is not quite the same as (6.3). The two differences are the 1 being added in the beginning of (6.10) and that there is an $G(t-s)$ in (6.10) and a $G(s)$ in (6.3).

Le's Question 6.4 Nick, I will go over your questions tomorrow morning... Talk to you soon.

7 Meeting on Nov. 19th

Le's Question 7.1 Try to decode the proof of Theorem 3.1 and see if we can prove this theorem using our language (in the Walsh framework instead of Da Prato's). You need to study the green book [21] for invariant measures.

8 Meeting on Nov. 26

Le's Question 8.1 This is related to Le's Question 6.3. The Bessel kernel function is a good example. Let's use the notion from wikipedia:

https://en.wikipedia.org/wiki/Bessel_potential

Then, let $f(x) = \frac{1}{(1+|x|^2)^{s/2}}$. Then the Fourier transform of f is given by G_s on the wikipedia. The condition (11) requires that

$$s > 2.$$

Let's take $s > d$, then we see that $G_s(x)$, which is $\gamma(x)$ in [39], is in $L^p(\mathbb{R}^d)$ for all p . However, in Hypothesis 2.1 of [39], one requires that

$$p \leq \frac{d}{d-2}.$$

Hence, we find examples that are allowed under our condition, namely (11), but not under Tessitore & Zabczyk [39].

Please work out the above details.

(Nick's Comments). I came up with something similar to what you just did but to the function $f(x) = \frac{1}{1 + |x|^\beta}$ which we were working on earlier. Like you said above, Hypothesis 2.1 requires that there exists a $p < \frac{d}{d-2}$ such that

$$\int \left(\frac{1}{1 + |x|^\beta} \right)^p dx < \infty \quad (20)$$

but I believe this means that (correct me if im wrong) we need

$$\beta \cdot p > d$$

We already agreed that under the Dalang condition we only require that $\beta > 2$ in order for

$$\int \frac{1}{1 + |x|^\beta} dx < \infty$$

So for a β sufficiently close to 2 and for a large d , we will not be able to find a $p < \frac{d}{d-2}$ such that $\beta \cdot p > d$. For example, when $d = 50$ and $\beta = 3$ then there does not exist $p < \frac{50}{48}$ so that (8.1) above holds. This proves that the Dalang condition is more general.

However, my counter exampal does not work for smaller d and more specifically for $d = 3$. □

Nick's Question 8.1 I believe what I have below is the whole question and set up in our language. As of now there seem to be five steps which I have addressed below that need to be completed.

First let $\rho(x)$ be an admissable weight as in [39] and consider the following SPDE.

$$\begin{cases} \frac{\partial u}{\partial t}(t, x) - \frac{1}{2} \Delta u(t, x) = b(x, u(t, x)) \dot{W}(t, x) & x \in \mathbb{R}^d, t > 0 \\ u(0, x) = \varphi(x) & \varphi(x) \in L^2_\rho(\mathbb{R}^d) \end{cases} \quad (21)$$

In (8.2), b satisfies some Lipschitz conditions to be determined and W is a Gaussian Processes that is white in time and colored in space as defined on page 51 of [18](the minicourse).

Step 1: Following [39] we first need to define a mild solution to (8.2). Im guessing the correct mild solution to define is

$$u(t, x) = \int_{\mathbb{R}^d} \varphi(y) P_t(x - y) dy + \int_0^t \int_{\mathbb{R}^d} b(s, u(s, y)) P_{t-s}(x - y) W(ds, dy) \quad (22)$$

where $P_t(x)$ is the standard heat kernel. For the followng, we denote the solution to (8.3) as $u(t, x, \varphi)$

Step 2: We need to show the existence and uniqueness to a solution of (8.3).

Step 3: We need to show that $u(t, \cdot, \varphi)$ is an object in $L^2_\rho(\mathbb{R}^d)$

Step 4: Can we show that the moment estimates in your paper [13] hold for the solution of (8.3)?

The above steps address all of the topics discussed in part 1 and part 2 of [39]. After verifying each of the above four steps we can start to address Theorem 3.1 in [39].

First, in [39] they assume that the solution to (8.2) is bounded in probability and we may or may not need to assume the same. Second, we define the law of our solution $u(t, \cdot, \varphi)$ as follows:

$$\mathcal{L}(u(t, \cdot, \varphi))(A) = \mathbb{P}(\omega \in \Omega : u(t, \cdot, \varphi) \in A) \quad A \in \mathcal{B}(L_\rho^2(\mathbb{R}^d)) \quad (23)$$

Then $\mathcal{L}(u(t, \cdot, \varphi))$ is a measure on $L_\rho^2(\mathbb{R}^d)$

Step 5: As in the proof of Theorem 3.1 in [39], if we can show that $\{\mathcal{L}(u(t, \cdot, \varphi))\}_{t>1}$ is tight in L_ρ^2 then the family of laws

$$\frac{1}{T} \int_1^{T+1} \mathcal{L}(u(t, \cdot, \varphi)) dt, \quad T > 0 \quad (24)$$

will also be tight and by (6.1.2) of [21], the weak limit of any converging sequence

$$\frac{1}{T_j} \int_1^{T_j+1} \mathcal{L}(u(t, \cdot, \varphi)) dt, \quad T_j \rightarrow \infty \quad (25)$$

is invariant for (8.3) in L_ρ^2 .

9 Discussions on Dec. 3rd

$$(F_\alpha f)(x) := \int_0^1 ds \int_{\mathbb{R}^d} dy (1-s)^{\alpha-1} G(1-s, x-y) f(s, y).$$

$$Y(s, y) = \int_0^s \int_{\mathbb{R}^d} (s-r)^{-\alpha} G(s-r, y-z) \rho(u(r, z)) W(dr, dz).$$

Then

$$\begin{aligned} (F_\alpha Y)(x) &= \int_0^1 ds \int_{\mathbb{R}^d} dy (1-s)^{\alpha-1} G(1-s, x-y) \\ &\quad \times \int_0^s \int_{\mathbb{R}^d} (s-r)^{-\alpha} G(s-r, y-z) \rho(u(r, z)) W(dr, dz) \\ &= \int_0^1 ds (1-s)^{\alpha-1} \int_0^s \int_{\mathbb{R}^d} (s-r)^{-\alpha} G(1-r, x-z) \rho(u(r, z)) W(dr, dz) \\ &= \int_0^1 \int_{\mathbb{R}^d} \left(\int_r^1 ds (1-s)^{\alpha-1} (s-r)^{-\alpha} \right) G(1-r, x-z) \rho(u(r, z)) W(dr, dz) \\ &= \frac{\pi}{\sin(\pi\alpha)} \int_0^1 \int_{\mathbb{R}^d} G(1-r, x-z) \rho(u(r, z)) W(dr, dz) \end{aligned}$$

As for the Step 2 of Theorem 3.1, we can do the following:

10 Discussions on Dec. 10th

Nick's Question 10.1 On page 275 of the paper, Tessitore defines the solution in equation (2.3) and the integrator is dW_s and right below it says that W has the identity operator as the covariance operator but this means that W is a white noise. In the

minicoursre is says that we can not deal with equations in higher dimensions that are forced by white noise so why can the integrator here a white noise when we are working with the heat equation on $\mathbb{R}^d$?

Nick's Question 10.2 This is something that I didn't notice until now, but in your question 6.2, you said you wanted to prove Theorem 3.3 with only the condition that

$$\int_{\mathbb{R}^d} \Gamma(dy) |y|^{2-d} < \infty$$

Did you mean Theorem 3.1?

Also im guessing we also need to assume Dalang's condition

$$\Upsilon(\beta) = \int_{\mathbb{R}^d} \frac{\hat{f}(d\xi)}{\beta + |\xi|^2} < \infty \quad \beta > 0$$

which is required by your paper [15] to have a solution and use the moment estimates which will show that our solution is almost surely in the weighted space, L_ρ^2 .

Nick's Question 10.3 (Showing step 2 from my question (8.1)) An issue we have is that we need to show that our solution is inside the weighted space so that the Law of the solution will be a measure in the L_ρ^2 space. There have been two ways that we discussed that we could show this:

(1)

$$E(\|u(t, \cdot)\|_\rho^2) = \int_{\mathbb{R}^d} \|u(t, x)\|_2^2 \rho(x) dx < \infty$$

(2)

$$\sup_{0 \leq t \leq T} E(\|u(t, \cdot)\|_\rho^2) = \sup_{0 \leq t \leq T} \int_{\mathbb{R}^d} \|u(t, x)\|_2^2 \rho(x) dx < \infty$$

First of all, in either (1) or (2) all we would have at best is that $u(t, \cdot) \in L_\rho^2$ almost surely and I can't tell from reading the Tessitore paper if that is good enough. For sure, in the Tessitore paper, their solution $X^x(t)(\omega) \in L_\rho^2$ for every $\omega \in \Omega$ and this is because of the way the stochastic integral is defined in the Da Prato textbook. Also, I'm not sure whether we need (1) or (2) but from the looks of the Tessitore paper it seems like (2) would be needed. That is because in the Tessitore paper we, have to work with

$$\frac{1}{T} \int_1^{T+1} \mathcal{L}(u(t, x)) dt$$

which is used to find the invariant measure and above the laws $\mathcal{L}(u(t, x))$ I believed are defined as

$$\mathcal{L}(u(t, x))(A) = \mathbb{P}(\omega \in \Omega : u(t, \cdot)(\omega) \in A \quad A \in \mathcal{B}(L_\rho^2))$$

There is an issue with even showing (1) from above. We need for $H(t, \gamma_2) < \infty$ from the moment bounds from your paper [15]. It seems the best we can do is Lemma 2.1 from [15] which says that

$$\limsup_{t \rightarrow \infty} \frac{1}{t} \log(H(t : \gamma)) \leq \inf\{\beta > 0 : \Upsilon(2\beta) < \frac{1}{2\gamma}\} \quad (26)$$

So first we would need to make sure that $\inf\{\beta > 0 : \Upsilon(2\beta) < \frac{1}{2\gamma}\} < \infty$ or in other words show that $\{\beta > 0 : \Upsilon(2\beta) < \frac{1}{2\gamma}\}$ is nonempty. This will restrict us to certain correlation functions $f(x)$. Another concern is that even if we assume the right hand side of (10.1) is finite, then all we know really is that infinitely often

$$H(t, \gamma) \leq \exp(Kt) \quad K = \text{RHS of (10.1)}$$

so there may still be some t such that $H(t, \gamma)$ is infinite and then this would cause a trouble in showing (2) from above.

Nick's Question 10.4 (Continue of question 10.3) I have looked through your paper [16] and have found more information about the behavior of $H(t; \gamma)$. However there seems to be a couple differences of the definitions of H from [16] and [15].

Definition from [15]: Let f be the correlation function the following functions:

$$k(t) := \int_{\mathbb{R}^d} f(z)G(t, z)dz = (2\pi)^{-d} \int_{\mathbb{R}^d} \hat{f}(d\xi) \exp\left(-\frac{t|\xi|^2}{2}\right) \quad (27)$$

$$h_0(t) := 1 \quad (28)$$

$$h_n(t) := \int_0^t ds h_{n-1}(s)k(t-s) \quad (29)$$

$$H(t; \gamma) := \sum_{n=0}^{\infty} \gamma^n h_n(t) \quad (30)$$

One question I have from (10.2) is that if $\hat{f}(d\xi) = f(\xi)d\xi$. I know we are using the identity $\int f\hat{g} = \int \hat{f}g$ to get the equality in (10.2).

Definition from [16]: Let f be the correlation function the following functions:

$$k(t) := \int_{\mathbb{R}^d} f(z)G(t, z)dz = (2\pi)^{-d} \int_{\mathbb{R}^d} \hat{f}(d\xi) \exp\left(-\frac{\nu t}{2}|\xi|^2\right) \quad (31)$$

$$T_\nu(t, x) := \exp\left(-\frac{|x|^2}{2\nu}\right) \quad (32)$$

$$h_0(t, y) := 1 \quad \text{for } t > 0 \text{ and } y \in \mathbb{R}^d \quad (33)$$

$$h_n(t, y) := \int_0^t ds h_{n-1}(s, y)k(t-s)T_{\nu/4}(t-s, y) \quad (34)$$

$$H_\nu(t, y; \gamma) := \sum_{n=0}^{\infty} \gamma^n h_n(t, y) \quad (35)$$

So above in [16] there is the addition of the T function and another parameter ν . Page 4 of [16] explains where the ν is coming from.

Aside from the differences in the definitions of H , [16] gives us further information about the finiteness of H as defined in [16]. We have the following Lemma on page 11.

Lemma 2.5 For all $t \geq 0$ and $\gamma \geq 0$,

$$H_\nu(t; \gamma) \leq \exp(\theta t) \quad (36)$$

where this constant θ can be chosen as

$$\theta := \theta(\nu, \gamma) = \inf\{\beta > 0 : \Upsilon(2\beta/\nu) < \frac{\nu}{2\gamma}\} \quad (37)$$

Moreover, if $\Upsilon(0) < \infty$ and $\gamma < 2\Upsilon(0)/\nu$ then,

$$H_\nu(t; \gamma) \leq \frac{\nu}{\nu - 2\gamma\Upsilon(0)} \quad \text{for all } t \geq 0 \quad (38)$$

After seeing Lemma 2.5 we have some necessary requirements to ensure the solution (8.3) of the SPDE (8.2) to be almost surely in L^2_ρ . Namely, we need that $\gamma_2 < 2\Upsilon(0)/\nu$ and that θ and $\Upsilon(0)$ from lemma 2.5 to be finite.

Details Showing the Solution of (8.2) is in L_ρ^2 .

We consider the SPDE

$$\begin{cases} \frac{\partial u}{\partial t}(t, x) - \frac{1}{2}\Delta u(t, x) = b(x, u(t, x))\dot{W}(t, x) & x \in \mathbb{R}^d, t > 0 \\ u(0, x) = \varphi(x) & \varphi(x) \in L_\rho^2(\mathbb{R}^d) \end{cases}$$

with mild solution given as

$$u(t, x) = \int_{\mathbb{R}^d} \varphi(y) G_t(x - y) dy + \int_0^t \int_{\mathbb{R}^d} b(s, u(s, y)) G_{t-s}(x - y) W(ds, dy)$$

We use the moment estimates from [15] to show that $u(t, x)$ is almost surely in L_ρ^2 but first there are assumptions that need to be made which are as follows:

- (a) In order for the solution to exist and be unique we must assume Dalang's condition, $\Upsilon(\beta) < \infty$ for some and therefore every $\beta > 0$
- (b) The initial condition needs to satisfy the integrability property (1.2) from [15] which states that

$$\int_{\mathbb{R}^d} e^{-a|x|^2} |\mu|(dx) < \infty$$

and I believe that this translates to

$$\int_{\mathbb{R}^d} e^{-a|x|^2} |\varphi(x)| dx < \infty$$

Instead, we may need to work under condition

$$\phi \in L_\rho^1(\mathbb{R}^d).$$

Because $\rho(x) \geq Ce^{-|x|^2} =: \rho_*(x)$ for all x , $L_\rho^1(\mathbb{R}^d) \subseteq L_{\rho_*}^1(\mathbb{R}^d)$.

- (c) Mentioned above we also need that θ and $\Upsilon(0)$ from lemma 2.5 to be finite and as well as $\gamma_2 < 2\Upsilon(0)/\nu$

Assuming the above we can now justify the following steps:

$$\mathbb{E} (\|u(t, \cdot)\|_\rho^2) = \mathbb{E} \int_{\mathbb{R}^d} |u(t, x)|^2 \rho(x) dx \tag{39}$$

$$\leq C_1 \cdot H(t, \gamma_2) \int_{\mathbb{R}^d} \left(1 + (|\varphi| * G(t, \cdot))(x)\right)^2 \rho(x) dx \tag{40}$$

$$\leq C_2 H(t, \gamma_2) \int_{\mathbb{R}^d} \left(1 + (|\varphi| * G(t, \cdot))^2(x)\right) \rho(x) dx \tag{41}$$

$$= C_2 H(t, \gamma_2) \int_{\mathbb{R}^d} \rho(x) + \left(\int_{\mathbb{R}^d} G_t(x - y) |\varphi(y)| dy\right)^2 \rho(x) dx \tag{42}$$

$$\leq C_3 H(t, \gamma_2) \int_{\mathbb{R}^d} \left(\int_{\mathbb{R}^d} G_t(x - y) |\varphi(y)| dy\right)^2 \rho(x) dx \tag{43}$$

$$\leq C_3 H(t, \gamma_2) \int_{\mathbb{R}^d} \int_{\mathbb{R}^d} G_t(x - y) |\varphi(y)|^2 \rho(x) dx dy \tag{44}$$

$$= C_3 H(t, \gamma_2) \int_{\mathbb{R}^d} (\rho * G_t)(y) |\varphi(y)|^2 dy \tag{45}$$

$$\leq C_3 H(t, \gamma_2) C(T) \int_{\mathbb{R}^d} \rho(y) |\varphi(y)|^2 dy \tag{46}$$

$$\leq \infty \quad t \in [0, T] \tag{47}$$

Since we assume (a)-(c) then Lemma 2.5 ensures that $H(t, \gamma_2)$ is finite for every t .

Le's Question 10.1 Can we allow the following weight functions:

$$\rho(x) = e^{-|x|^\alpha}, \quad \alpha \in (1, 2)?$$

Can we make such ρ in order that

$$G(t, \cdot) * \rho \leq C_\rho(T)\rho?$$

□

11 Discussions on Jan. 16th

Nick's Question 11.1 (Compactness of the F_α operator) The reason for compactness stated in the paper on page 277 for the F_α operator should transfer to the F_α operator defined as follows.

$$(F_\alpha f)(x) := \int_0^1 ds \int_{\mathbb{R}^d} dy (1-s)^{\alpha-1} G(1-s, x-y) f(s, y)$$

The compactness of this operator relies solely on proposition 2.1 which has no relation to hypothesis 2.1 (i).

Nick's Question 11.2 (Step 2 details)

$$Y^x(s) = \int_0^s (s-\varsigma)^{-\alpha} S(s-\varsigma) M(B(X^x(\varsigma))) dW_\varsigma$$

Below, the constant K changes to satisfy certain inequalities. This shouldn't cause too much confusion.

The main claim in this step is that

$$E \int_0^1 |Y^x(s)|_\rho^q ds \leq k_1 |x|_\rho^q \quad (48)$$

but there are a lot of details skipped in showing this. First, let's just assume that all of the claims on the bottom of page 277 are justified, namely the last centered equation which says that

$$E \int_0^1 |Y^x(s)|_\rho^q ds \leq K \cdot E \int_0^1 |X^x(\varsigma)|_\rho^q d\varsigma \quad (49)$$

Then according to the top of page 278 we can conclude (11.1) by using the result (not proven in this paper) that

$$E |X^x(t)|_\rho^q \leq k_2 (1 + |x|_\rho^q) \quad (50)$$

. But if we use (11.3) then all we can say is that

$$E \int_0^1 |Y^x(s)|_{\hat{\rho}}^q ds \leq K \cdot E \int_0^1 |X^x(\varsigma)|_{\hat{\rho}}^q d\varsigma \quad (51)$$

$$\leq K \int_0^1 1 + |x|_{\hat{\rho}}^q d\varsigma \quad (52)$$

$$= K(1 + |x|_{\hat{\rho}}^q) \quad (53)$$

This is an issue since all this doesn't prove (11.1). However if $|x|_{\hat{\rho}} \neq 0$ then we could say (11.1) is true and it seems safe to say that $|x|_{\hat{\rho}} \neq 0$ so I don't think this really is an issue.

Now going back to (11.2), we have the following.

$$E \int_0^1 |Y^x(s)|_{\hat{\rho}}^q ds = E \int_0^1 \left| \int_0^s (s - \varsigma)^{-\alpha} S(s - \varsigma) M(B(X^x(\varsigma))) dW_{\varsigma} \right|_{\hat{\rho}}^q ds \quad (54)$$

$$\leq E \int_0^1 \left(\int_0^s (s - \varsigma)^{-2\alpha} \|S(s - \varsigma) M(B(X^x(\varsigma)))\|_{L^2}^2 d\varsigma \right)^{q/2} ds \quad (55)$$

$$\leq K \cdot E \int_0^1 \left(\int_0^s (s - \varsigma)^{-2\alpha - \sigma} |B(X^x(\varsigma))|_{\hat{\rho}}^2 d\varsigma \right)^{q/2} ds \quad (56)$$

In (11.9) we used proposition 2.4. Now we can use youngs inequality for convolutions which says that

$$\|f * g\|_r \leq \|f\|_p \|g\|_q$$

where $1 \leq p, q \leq r < \infty$ and $\frac{1}{p} + \frac{1}{q} = \frac{1}{r} + 1$. To do this we say that

$$f(t) = t^{-2\alpha + \sigma}$$

and

$$g(t) = |B(X^x(t))|_{\hat{\rho}}^2$$

and lastly let $p = 1$, $q = \frac{q}{2}$ and $r = \frac{q}{2}$. We also need to multiply $f(t)$ by something like $1_{[0, \infty)}(t)$ since the bounds on the inner integral in (11.9) are from 0 to s instead of $-\infty$ to ∞ . After applying Youngs inequality on (11.9) we are left with the following

$$K \cdot E \int_0^1 \left(\int_0^s (s - \varsigma)^{-2\alpha - \sigma} |B(X^x(\varsigma))|_{\hat{\rho}}^2 d\varsigma \right)^{q/2} ds \leq K \cdot E \left(\int_0^1 t^{-2\alpha - \sigma} dt \right)^{q/2} \int_0^1 |B(X^x(t))|_{\hat{\rho}}^q dt \quad (57)$$

$$= K \cdot (1 - 2\alpha - \sigma)^{-q/2} E \int_0^1 |B(X^x(t))|_{\hat{\rho}}^q dt \quad (58)$$

At this point the paper skips a lot of steps. As you can see from the bottom centered equation on page 277, the paper claims that

$$|B(X^x(t))|_{\hat{\rho}}^q \leq l^q |\hat{\rho}|_{L^1}^q |X^x(t)|_{\hat{\rho}}^q$$

and this not necessarily true. Usig the Lipschitz property of B , what we can say is that

$$|B(X^x(t))|_{\hat{\rho}}^q \leq (|B(0)|_{\hat{\rho}} + l|X^x(t)|_{\hat{\rho}})^q \quad (59)$$

$$\leq 2^{q-1} \left(|B(0)|_{\hat{\rho}}^q + l^q |X^x(t)|_{\hat{\rho}}^q \right) \quad (60)$$

$$\leq 2^{q-1} \left(l_0^q |\hat{\rho}|_{L^1}^q + l^q |X^x(t)|_{\hat{\rho}}^q \right) \quad (61)$$

$$\leq K(1 + |X^x(t)|_{\hat{\rho}}^q) \quad (62)$$

Now plugging (11.15) into (11.11) we see that

$$K \cdot (1 - 2\alpha - \sigma)^{-q/2} E \int_0^1 |B(X^x(t))|_{\hat{\rho}}^q dt \leq K \cdot (1 - 2\alpha - \sigma)^{-q/2} E \int_0^1 1 + |X^x(t)|_{\hat{\rho}}^q dt \quad (63)$$

$$\leq K \left(1 + E \int_0^1 |X^x(t)|_{\hat{\rho}}^q dt \right) \quad (64)$$

And as before if

$$E \int_0^1 |X^x(t)|_{\hat{\rho}}^q dt \neq 0$$

then we may say that

$$K \left(1 + E \int_0^1 |X^x(t)|_{\hat{\rho}}^q dt \right) \leq K E \int_0^1 |X^x(t)|_{\hat{\rho}}^q dt$$

and it makes sense that

$$E \int_0^1 |X^x(t)|_{\hat{\rho}}^q dt \neq 0$$

so maybe this isn't an issue.

Nick's Question 11.3 (Clarifiation of Spectral Density vs Correlation function)

I might have asked this before but I just wanted to clarify the difference between the correlation function and spectral density how it is defined in the mini course and how it is also referred to in the paper [39] as well in your paper [15].

First in the mini-course we start with a non-negative and non-negative definite tempered measure $\Gamma(dx)$ which means that

$$\int_{\mathbb{R}^d} \Gamma(dx) (\varphi * \tilde{\varphi})(x) \geq 0 \quad (65)$$

We also require that

$$\int_{\mathbb{R}^d} \frac{\Gamma(dx)}{(1 + |x|^2)^r} < \infty \quad (66)$$

Now, define

$$\{F(\varphi) : \varphi \in C_0^\infty(\mathbb{R}^{d+1})\}$$

to be a gaussian process that is white in time and colured in space where

$$E(F(\varphi)F(\psi)) = J(\varphi, \psi) = \int_{\mathbb{R}_+} ds \int_{\mathbb{R}^d} dx \int_{\mathbb{R}^d} dy \varphi(s, x) f(x - y) \psi(s, y)$$

whenever $\Gamma(dx) = f(x)dx$. We informally say that

$$E(\dot{F}(t, x)\dot{F}(s, y)) = \delta_0(t - s)f(x - y)$$

and above we call f the correlation function. Now by Bochner's Theorem, there exists a measure μ such that $\Gamma(dx) = \mathcal{F}\mu(dx)$ which means that

$$\int \Gamma(dx)\varphi(x) = \int \mu(dx)\mathcal{F}\varphi(x)$$

Now, the mini course calls $\mu(dx)$ the spectral measure and when this has density, say, $\mu(dx) = s(x)dx$ then $s(x)$ is the spectral density.

Now this seems to be a little different from your paper [15] and also your paper [16]. In those papers you say that $\hat{f}(x) = \int e^{-ix\xi}f(\xi)d\xi$ is called the spectral measure. However this seems a little off. From above we know that

$$\Gamma(dx) = f(x)dx = \mathcal{F}\mu(dx) = \mathcal{F}s(x)dx$$

and we call $\mu(dx) = s(x)dx$ the spectral measure not $\hat{f}$.

Question 1

With all this being said, in the paper [39], does the $\gamma(x)$, which is called the spectral density of Γ , from hypothesis 2.1(i) refer to $s(x)$ above and not $f(x)$? Our goal was to show that Dalang's condition

$$\int \frac{\hat{f}(dx)}{\beta + |x|^2} < \infty$$

could replace hypothesis 2.1(i) but Dalangs condition is on the correlation function and hypothesis 2.1(i) is about the spectral density. The two are related but not the same thing.

Question 2

This all goes back to "Le Question 6.2" where you say that you want to prove the same result from [39] except under the condition that

$$\int \Gamma(dy)|y|^{2-d} = \int f(y)|y|^{2-d}dy < \infty$$

but why do we want this to be the condition? This is a little different from Dalang's condition which says that

$$\int \frac{\hat{f}(dy)}{\beta + |y|^2} < \infty$$

Le's Question 11.1
called 2020-01-16.pdf.

Hi, Nick, I have uploaded that page on overleaf. The file is

Nick's Question 11.4

This is something I just realized but I feel like line (11.8) above needs some justification. This is almost the BDG-inequality which is given in the Green Book on page 58 however here we have $E|\int \dots|_p^q$ instead of $E|\int \dots|^q$.

Nick's Question 11.5 Is there a factorization formula for the Dalang language like in the Red Book on page 58? If not then if we want to follow the same proof outline as in the Tessitore paper then we would need to prove one.

A Possible Step 2 Inequality

The step two inequality is only place where the paper really relies on hypothesis 2.1 (i). Aside from this inequality, we would also need to prove a similar factorization formula which is what step 3 relies on. Below in Claim 1 is a possible step 2 inequality but there are a few steps that may be incorrect.

Claim 1: Let $\varphi \in L^2_{\hat{\rho}}$ be the initial condition. Then we have the following

$$E \int_0^1 |Y^\alpha(s, y)|^2_{\hat{\rho}} ds \leq \text{const.} (1 + |\varphi|^2_{\hat{\rho}})$$

where

$$Y^\alpha(s, y) = \int_0^s \int_{\mathbb{R}^d} (s - r)^{-\alpha} G(s - r, y - z) b(u(r, z)) W(dr, dz)$$

Proof:

$$\begin{aligned}
E \int_0^1 |Y^\alpha(s, y)|_{\hat{\rho}}^2 ds &= \int_0^1 ds \int_{\mathbb{R}^d} dy E |Y^\alpha(s, y)|^2 \hat{\rho}(y) \\
&= \int_{\mathbb{R}^d} dy \hat{\rho}(y) \int_0^1 ds E |Y^\alpha(s, y)|^2 \\
&\leq \int_{\mathbb{R}^d} dy \hat{\rho}(y) \int_0^1 ds E \left| \int_0^s dr \int_{\mathbb{R}^d} dz \int_{\mathbb{R}^d} d\xi (s-r)^{-\alpha} G_{s-r}(y-z) \right. \\
&\quad \left. \times b(u(r, z)) f(z-\xi) (s-r)^{-\alpha} G_{s-r}(y-\xi) b(u(r, \xi)) \right| \\
&\leq \int_{\mathbb{R}^d} dy \hat{\rho}(y) \int_0^1 ds E \int_0^s dr \int_{\mathbb{R}^d} dz \int_{\mathbb{R}^d} d\xi (s-r)^{-\alpha} G_{s-r}(y-z) \\
&\quad \times \left| b(u(r, z)) \right| f(z-\xi) (s-r)^{-\alpha} G_{s-r}(y-\xi) \left| b(u(r, \xi)) \right| \\
&= \int_{\mathbb{R}^d} dy \hat{\rho}(y) \int_0^1 ds \int_0^s dr \int_{\mathbb{R}^d} dz \int_{\mathbb{R}^d} d\xi (s-r)^{-\alpha} G_{s-r}(y-z) \\
&\quad \times f(z-\xi) (s-r)^{-\alpha} G_{s-r}(y-\xi) E \left(\left| b(u(r, \xi)) \right| \left| b(u(r, z)) \right| \right) \\
&= \int_{\mathbb{R}^d} dy \hat{\rho}(y) \int_0^1 ds \int_0^s dr \int_{\mathbb{R}^d} dz \int_{\mathbb{R}^d} d\xi (s-r)^{-\alpha} G_{s-r}(z) \\
&\quad \times f(z-\xi) (s-r)^{-\alpha} G_{s-r}(\xi) E \left(\left| b(u(r, y-\xi)) \right| \left| b(u(r, y-z)) \right| \right) \\
&\leq K_1 \int_{\mathbb{R}^d} dy \hat{\rho}(y) \int_0^1 ds \int_0^s dr \int_{\mathbb{R}^d} dz \int_{\mathbb{R}^d} d\xi (s-r)^{-\alpha} G_{s-r}(z) \\
&\quad \times f(z-\xi) (s-r)^{-\alpha} G_{s-r}(\xi) E \left(\left(1 + |u(r, y-\xi)| \right) \left(1 + |u(r, y-z)| \right) \right) \\
&\leq K_1 \int_{\mathbb{R}^d} dy \hat{\rho}(y) \int_0^1 ds \int_0^s dr \int_{\mathbb{R}^d} dz \int_{\mathbb{R}^d} d\xi (s-r)^{-2\alpha} G_{s-r}(z) \\
&\quad \times f(z-\xi) G_{s-r}(\xi) \sqrt{E \left(1 + |u(r, y-\xi)| \right)^2} \sqrt{E \left(1 + |u(r, y-z)| \right)^2} \\
&\leq K_1 \int_{\mathbb{R}^d} dy \hat{\rho}(y) \sup_{x \in \mathbb{R}^d} E \left(1 + |u(r, y-x)| \right)^2 \int_0^1 ds \int_0^s dr (s-r)^{-2\alpha} \\
&\quad \underbrace{\int_{\mathbb{R}^d} dz \int_{\mathbb{R}^d} d\xi G_{s-r}(z) f(z-\xi) G_{s-r}(\xi)}_{\text{finite by page 57 of mini course}} \\
&\leq K_2 \int_{\mathbb{R}^d} dy \hat{\rho}(y) \sup_{x \in \mathbb{R}^d} E \left(1 + |u(r, y-x)| \right)^2 \underbrace{\int_0^1 ds \int_0^s dr (s-r)^{-2\alpha}}_{\text{finite when } \alpha < 1/2} \\
&\leq K_3 \int_{\mathbb{R}^d} dy \hat{\rho}(y) \sup_{x \in \mathbb{R}^d} E \left(1 + |u(r, y-x)| \right)^2 \\
&\leq K_4 \int_{\mathbb{R}^d} dy \hat{\rho}(y) \sup_{x \in \mathbb{R}^d} E \left(1 + |u(r, y-x)|^2 \right) \\
&= K_4 \sup_{x \in \mathbb{R}^d} E \int_{\mathbb{R}^d} dy \left(\hat{\rho}(y) + \hat{\rho}(y) |u(r, y-x)|^2 \right) \quad \text{switching sup and } \int \text{ is troublesome} \\
&= K_4 \sup_{x \in \mathbb{R}^d} \int_{\mathbb{R}^d} \hat{\rho}(y) dy + E |u(r, \cdot - x)|_{\hat{\rho}(y)}^2 \\
&\leq K_4 \sup_{x \in \mathbb{R}^d} \left(\int_{\mathbb{R}^d} \hat{\rho}(y) dy + \frac{20}{k} (1 + |\varphi|_{\hat{\rho}}^2) \right) \quad \text{Inequality top of page 278, Tessitore paper. Not sure if this is possible since the} \\
&\quad \text{inequality is for } E |u(r, \cdot)|^2
\end{aligned}$$

Le's Question 11.2

Here is what I want to try: For any $q \geq 2$, we have by the BDG-inequality,

$$\begin{aligned} \|Y^\alpha(s, y)\|_q^2 \leq & C_q \int_0^s dr (s-r)^{-2\alpha} \iint_{\mathbb{R}^{2d}} dz_1 dz_2 G(s-r, y-z_1) \|b(u(r, z_1))\|_q f(z_1 - z_2) \\ & \times G(s-r, y-z_2) \|b(u(r, z_2))\|_q \end{aligned}$$

For simplicity, let's assume that $|b(u)| \leq \text{Lip}_b |u|$. By the moment bounds from [15], we have that

$$\|b(u(r, z_i))\|_q \leq c_q \text{Lip}_b J_0(r, z_i), \quad i = 1, 2.$$

Hence,

$$\begin{aligned} \|Y^\alpha(s, y)\|_q^2 \leq & C_q c_q^2 \text{Lip}_b^2 \int_0^s dr (s-r)^{-2\alpha} \\ & \times \iint_{\mathbb{R}^{2d}} dz_1 dz_2 G(s-r, y-z_1) G(s-r, y-z_2) f(z_1 - z_2) \\ & \times \iint_{\mathbb{R}^{2d}} \mu(dw_1) \mu(dw_2) G(r, z_1 - w_1) G(r, z_2 - w_2) \end{aligned}$$

Notice that, for $i = 1, 2$, it holds that

$$G(s-r, y-z_i) G(r, z_i - w_i) = G(s, y-w_i) G\left(\frac{r(s-r)}{s}, z_i - w_i - \frac{r}{s}(y-w_i)\right). \quad (67)$$

Therefore,

$$\begin{aligned} \|Y^\alpha(s, y)\|_q^2 \leq & C_q c_q^2 \text{Lip}_b^2 \int_0^s dr (s-r)^{-2\alpha} \\ & \times \iint_{\mathbb{R}^{2d}} \mu(dw_1) \mu(dw_2) G(r, y-w_1) G(r, y-w_2) \\ & \times \iint_{\mathbb{R}^{2d}} dz_1 dz_2 f(z_1 - z_2) \\ & \times \prod_{i=1}^2 G\left(\frac{r(s-r)}{s}, z_i - w_i - \frac{r}{s}(y-w_i)\right). \end{aligned}$$

By the Fourier transform, we see that

$$\begin{aligned} \iint_{\mathbb{R}^{2d}} dz_1 dz_2 f(z_1 - z_2) \prod_{i=1}^2 G\left(\frac{r(s-r)}{s}, z_i - w_i - \frac{r}{s}(y-w_i)\right) \\ \leq (2\pi)^{-d} \int_{\mathbb{R}^d} \hat{f}(d\xi) \exp\left(-\frac{r(s-r)}{s} |\xi|^2\right). \quad (68) \end{aligned}$$

Therefore,

$$\begin{aligned} \|Y^\alpha(s, y)\|_q^2 \leq & (2\pi)^{-d} C_q c_q^2 \text{Lip}_b^2 \int_0^s dr (s-r)^{-2\alpha} \\ & \times \int_{\mathbb{R}^d} \hat{f}(d\xi) \exp\left(-\frac{r(s-r)}{s} |\xi|^2\right) \\ & \times \iint_{\mathbb{R}^{2d}} \mu(dw_1) \mu(dw_2) G(r, y-w_1) G(r, y-w_2). \end{aligned}$$

Then, when we take into account of the weight function $\hat{\rho}(\cdot)$, we have

$$\int_{\mathbb{R}^d} \|Y^\alpha(s, y)\|_q^2 \hat{\rho}(y) dy \leq (2\pi)^{-d} C_q c_q^2 \text{Lip}_b^2 \int_0^s dr (s-r)^{-2\alpha} \quad (69)$$

$$\times \int_{\mathbb{R}^d} \hat{f}(d\xi) \exp\left(-\frac{r(s-r)}{s} |\xi|^2\right) \quad (70)$$

$$\times \int_{\mathbb{R}^d} dy \hat{\rho}(y) \iint_{\mathbb{R}^{2d}} \mu(dw_1) \mu(dw_2) G(r, y-w_1) G(r, y-w_2). \quad (71)$$

It reduce to study the following function near zero:

$$r \mapsto \int_{\mathbb{R}^d} dy \hat{\rho}(y) \iint_{\mathbb{R}^{2d}} \mu(dw_1) \mu(dw_2) G(r, y-w_1) G(r, y-w_2) =: Q(r).$$

If $\mu(dx) = dx$, then $Q(r) = \|\hat{\rho}\|_{L^1}$, which is a constant function in r . When $\mu = \delta_0$, then

$$Q(r) = \int_{\mathbb{R}^d} G(r, y)^2 \hat{\rho}(y) dy \approx \frac{C}{r^{d/2}} \quad \text{as } r \downarrow 0.$$

Suppose that we do assume that the initial data μ is a function, then $Q(r)$ will not blow up as $r \downarrow 0$. Hence, for some constant depending on the initial data K_ρ

$$\int_{\mathbb{R}^d} \|Y^\alpha(s, y)\|_q^2 \hat{\rho}(y) dy \leq (2\pi)^{-d} K_\rho C_q c_q^2 \text{Lip}_b^2 \int_0^s dr (s-r)^{-2\alpha} \quad (72)$$

$$\times \int_{\mathbb{R}^d} \hat{f}(d\xi) \exp\left(-\frac{r(s-r)}{s} |\xi|^2\right) \quad (73)$$

Now the finiteness of the above integral gives the condition on f . Therefore,

$$Y^\alpha \in L^q((0, 1), L_\rho^2).$$

Here K_ρ is somehow $\|\mu\|_\rho^2$. Once you can show that, we are done.

Step 2 Inequality: Let $\varphi \in L_\rho^2$ be the initial condition then for $q > 2$

$$E \int_0^1 |Y(s, y)|_\rho^q ds \leq K |\varphi|_\rho^q \quad (74)$$

for some constant K .

Proof: In the following proof, all constants get absorbed into K so K will change from line to

line. The following continues from equation (11.22)

$$\begin{aligned}
\underbrace{\int_{\mathbb{R}^d} \|Y(s, y)\|_q^2 \rho(y) dy}_{= \left\| \|Y(s, y)\|_q \right\|_\rho^2} &\leq K \int_0^s dr (s-r)^{-2\alpha} \int_{\mathbb{R}^d} \hat{f}(d\xi) \exp\left(-\frac{r(s-r)}{s} |\xi|^2\right) \\
&\quad \times \int_{\mathbb{R}^d} dy \rho(y) \iint_{\mathbb{R}^{2d}} \mu(dw_1) \mu(dw_2) G(r, y-w_1) G(r, y-w_2) \\
&= K \int_0^s dr (s-r)^{-2\alpha} \int_{\mathbb{R}^d} \hat{f}(d\xi) \exp\left(-\frac{r(s-r)}{s} |\xi|^2\right) \\
&\quad \times \int_{\mathbb{R}^d} dy \rho(y) \left(\int_{\mathbb{R}^d} dw G(r, y-w) |\varphi(w)| \right)^2 \\
&\leq K \int_0^s dr (s-r)^{-2\alpha} \int_{\mathbb{R}^d} \hat{f}(d\xi) \exp\left(-\frac{r(s-r)}{s} |\xi|^2\right) \\
&\quad \times \int_{\mathbb{R}^d} dy \rho(y) \int_{\mathbb{R}^d} dw G(r, y-w) |\varphi(w)|^2 \underbrace{\int_{\mathbb{R}^d} dw G(r, y-w)}_{\text{integral of heat kernel} = 1} \quad \text{by holders inequality} \\
&= K \int_0^s dr (s-r)^{-2\alpha} \int_{\mathbb{R}^d} \hat{f}(d\xi) \exp\left(-\frac{r(s-r)}{s} |\xi|^2\right) \\
&\quad \times \int_{\mathbb{R}^d} dw |\varphi(w)|^2 \int_{\mathbb{R}^d} dy G(r, y-w) \rho(y) \\
&\leq K \int_0^s dr (s-r)^{-2\alpha} \int_{\mathbb{R}^d} \hat{f}(d\xi) \exp\left(-\frac{r(s-r)}{s} |\xi|^2\right) \\
&\quad \times \int_{\mathbb{R}^d} dw |\varphi(w)|^2 C_\rho(T) \rho(w) \quad \text{admissible weight property} \\
&= C(T) K |\varphi|_\rho^2 \underbrace{\int_0^s dr (s-r)^{-2\alpha} \int_{\mathbb{R}^d} \hat{f}(d\xi) \exp\left(-\frac{r(s-r)}{s} |\xi|^2\right)}_{\text{assumption on } f \text{ makes this finite}} \\
&= K |\varphi|_\rho^2 \quad \text{for every } s \in [0, T] \text{ by the admissible weight property}
\end{aligned}$$

The above implies that

$$\left\| \|Y(s, y)\|_q \right\|_\rho \leq K |\varphi|_\rho \quad (75)$$

Now I will use (11.28) to prove (11.27)

$$\begin{aligned}
E|Y(s, y)|_\rho^q &= \int_{\Omega} dP \left(\int_{\mathbb{R}^d} dy Y(s, y)^2 \rho(y) \right)^{q/2} \\
&\leq \left(\int_{\mathbb{R}^d} dy \left(\int_{\Omega} dP Y(s, y)^q \rho(y)^{q/2} \right)^{2/q} \right)^{q/2} \quad \text{minkowski integral inequality} \\
&= \left(\int_{\mathbb{R}^d} dy \rho(y) \|Y(s, y)\|_q^2 \right)^{q/2} \\
&= \left\| \|Y(s, y)\|_q \right\|_\rho^q \\
&\leq K |\varphi|_\rho^q \quad \text{by (11.28)}
\end{aligned}$$

Now by integrating out s from $0 \rightarrow 1$ we are able to obtain the inequality in (11.27)

$$E \int_0^1 |Y(s, y)|_\rho^q ds \leq \int_0^1 K |\varphi|_\rho^q ds = K |\varphi|_\rho^q \quad (76)$$

□

The Factorization Formula

We want to show that the following factorization formula holds true.

$$\int_0^t \int_{\mathbb{R}^d} G_{t-s}(x-y)b(u(s,y))W(ds,dy) = \frac{\sin(\alpha\pi)}{\pi} \int_0^t (t-s)^{\alpha-1} G_{t-s} Y^\alpha(s) ds \quad (77)$$

where

$$Y^\alpha(s,y) = \int_0^s \int_{\mathbb{R}^d} (s-r)^{-\alpha} G(s-r,y-z)b(u(r,y))W(dr,dy)$$

This factorization is key for the proof on the tessitore paper to work. In the Da Pratos Green book on page 58 we have the following factorization

$$\int_0^t S(t-s)\phi(s)dW_s = \frac{\sin(\alpha\pi)}{\pi} \int_0^t (t-s)^{\alpha-1} Y_\alpha(s) ds \quad (78)$$

where

$$Y_\alpha(s) = \int_0^s (s-r)^{-\alpha} S(s-r)\phi(r)dW_r$$

and above $\phi(t) \in L_2(U, H)$, which are the Hilbert-Schmidt operators from U to H , and W_t is H valued.

Nick's Question 11.6 How do you prove (11.21)? It says in the Green Textbook that it is an easy application of the stochastic Fubini's theorem but I couldn't see how to do it.

Le's Question 11.3 We also have the stochastic Fubini's theorem, see Mini Course p.27 Theorem 5.30. You need to verify that we indeed can apply that theorem. Once that is done, the stochastic Fubini's theorem shows that

$$\begin{aligned} & \frac{\sin(\alpha\pi)}{\pi} \int_0^t (t-s)^{\alpha-1} [G_{t-s} Y^\alpha(s)](x) ds \\ &= \frac{\sin(\alpha\pi)}{\pi} \int_0^t ds (t-s)^{\alpha-1} \int_{\mathbb{R}^d} dy G_{t-s}(x-y) \int_0^s \int_{\mathbb{R}^d} (s-r)^{-\alpha} G(s-r,y-z)b(u(r,z))W(dr,dz) \\ &= \frac{\sin(\alpha\pi)}{\pi} \int_0^t ds (t-s)^{\alpha-1} \int_0^s \int_{\mathbb{R}^d} (s-r)^{-\alpha} G(t-r,x-z)b(u(r,z))W(dr,dz) \\ &= \frac{\sin(\alpha\pi)}{\pi} \int_0^t \int_{\mathbb{R}^d} W(dr,dz) G(t-r,x-z)b(u(r,z)) \int_r^t ds (s-r)^{-\alpha} (t-s)^{\alpha-1} \\ &= \int_0^t \int_{\mathbb{R}^d} G(t-r,x-z)b(u(r,z))W(dr,dz) \end{aligned}$$

where the last step is true provided $\alpha \in (0,1)$. This is the desired formula in Dalang's language.

Possible Justification of the Above Use of Fubini's Theorem: Fubini's Theorem is given in the Walsh Notes on page 296. In your question 11.3, Fubini's Theorem is applied in two different spots.

First we use it here,

$$\begin{aligned} & \frac{\sin(\alpha\pi)}{\pi} \int_0^t ds (t-s)^{\alpha-1} \underbrace{\int_{\mathbb{R}^d} dy G_{t-s}(x-y) \int_0^s \int_{\mathbb{R}^d} (s-r)^{-\alpha} G(s-r, y-z) b(u(r, z)) W(dr, dz)}_{\int_{\mathbb{R}^d} \mu(dy) \int_0^s \int_{\mathbb{R}^d} f(r, z, \omega, y) W(dr, dz)} \\ &= \frac{\sin(\alpha\pi)}{\pi} \int_0^t ds (t-s)^{\alpha-1} \int_0^s \int_{\mathbb{R}^d} (s-r)^{-\alpha} G(t-r, x-z) b(u(r, z)) W(dr, dz) \end{aligned}$$

Second we use it here,

$$\begin{aligned} & \frac{\sin(\alpha\pi)}{\pi} \int_0^t ds (t-s)^{\alpha-1} \int_0^s \int_{\mathbb{R}^d} (s-r)^{-\alpha} G(t-r, x-z) b(u(r, z)) W(dr, dz) \\ &= \frac{\sin(\alpha\pi)}{\pi} \int_0^t \int_{\mathbb{R}^d} W(dr, dz) G(t-r, x-z) b(u(r, z)) \int_r^t ds (s-r)^{-\alpha} (t-s)^{\alpha-1} \end{aligned}$$

For the first case: According to the Walsh notes, what we need to show is that

$$\begin{aligned} E \left[\int_{\mathbb{R}^d} dy G_{t-s}(x-y) \int_0^s dr (s-r)^{-2\alpha} \right. \\ \left. \times \iint_{\mathbb{R}^{2d}} dz_1 dz_2 G_{s-r}(y-z_1) G_{s-r}(y-z_2) \left| b(u(r, z_1)) b(u(r, z_2)) \right| f(z_1 - z_2) \right] < \infty \end{aligned} \quad (79)$$

We can show using the same techniques used in "Le's Question 11.2" and also in the proof of (11.27) to see that the integral in (11.32) is bounded above by

$$\begin{aligned} (11.32) &\leq \int_{\mathbb{R}^d} dy G_{t-s}(x-y) \int_0^s dr (s-r)^{-2\alpha} \int_{\mathbb{R}^d} \hat{f}(d\xi) \exp\left(-\frac{r(s-r)}{s} |\xi|^2\right) \int_{\mathbb{R}^d} dw G_r(y-w) |\varphi(w)|^2 \\ &= \int_0^s dr (s-r)^{-2\alpha} \int_{\mathbb{R}^d} \hat{f}(d\xi) \exp\left(-\frac{r(s-r)}{s} |\xi|^2\right) \int_{\mathbb{R}^d} dy \int_{\mathbb{R}^d} dw G_r(y-w) G_{t-s}(x-y) |\varphi(w)|^2 \\ &= \int_0^s dr (s-r)^{-2\alpha} \int_{\mathbb{R}^d} \hat{f}(d\xi) \exp\left(-\frac{r(s-r)}{s} |\xi|^2\right) \underbrace{\int_{\mathbb{R}^d} dw G_{t-(s-r)}(x-w) |\varphi(w)|^2}_{a(r) \quad r \in [0, s]} \quad (80) \end{aligned}$$

Since $a(r)$ is a continuous function on $[0, s]$ then it is bounded by some constant, M . So we see that (11.33) is bounded by

$$(11.33) \leq M \underbrace{\int_0^s dr (s-r)^{-2\alpha} \int_{\mathbb{R}^d} \hat{f}(d\xi) \exp\left(-\frac{r(s-r)}{s} |\xi|^2\right)}_{\text{finite by assumption}} \quad (81)$$

And this verifies (11.32) so the first appication of Fubini's Theorem is ok.

For the second case: For this case we need to show that

$$\begin{aligned} E \left[\int_0^t ds (t-s)^{\alpha-1} \int_0^s dr (s-r)^{-2\alpha} \right. \\ \left. \times \iint_{\mathbb{R}^{2d}} dz_1 dz_2 G(t-r, x-z_1) G(t-r, x-z_2) \left| b(u(r, z_1)) b(u(r, z_2)) \right| f(z_1 - z_2) \right] < \infty \end{aligned} \quad (82)$$

We again can pass the expectation through the integral, apply the Lipschitz property of b , apply Hölder's inequality, and then the moments estimates to see that

$$\begin{aligned}
(11.35) &\leq \int_0^t ds (t-s)^{\alpha-1} \int_0^s dr (s-r)^{-2\alpha} \\
&\quad \times \iint_{\mathbb{R}^{2d}} dz_1 dz_2 G(t-r, x-z_1) G(t-r, x-z_2) \|u(r, z_1)\|_2 \|u(r, z_2)\|_2 f(z_1 - z_2) \\
&\leq \int_0^t ds (t-s)^{\alpha-1} \int_0^s dr (s-r)^{-2\alpha} \iint_{\mathbb{R}^{2d}} dw_1 dw_2 G_t(x-w_1) G_t(x-w_2) |\varphi(w_1)| |\varphi(w_2)| \\
&\quad \times \iint_{\mathbb{R}^{2d}} dz_1 dz_2 f(z_1 - z_2) \prod_{i=1}^2 G\left(-\frac{r(t-r)}{t}, z_i - w_i - \frac{r}{s}(x - w_i)\right) \\
&\leq \int_0^t ds (t-s)^{\alpha-1} \int_0^s dr (s-r)^{-2\alpha} \iint_{\mathbb{R}^{2d}} dw_1 dw_2 G_t(x-w_1) G_t(x-w_2) |\varphi(w_1)| |\varphi(w_2)| \\
&\quad \times \int_{\mathbb{R}^d} \hat{f}(d\xi) \exp\left(-\frac{r(t-r)}{t} |\xi|^2\right) \\
&= J_0^2(t, x) \int_0^t ds (t-s)^{\alpha-1} \int_0^s dr (s-r)^{-2\alpha} \int_{\mathbb{R}^d} \hat{f}(d\xi) \exp\left(-\frac{r(t-r)}{t} |\xi|^2\right) \quad (83) \\
&= \int_0^t ds (t-s)^{\alpha-1} \int_0^s dr (s-r)^{-2\alpha} \int_{\mathbb{R}^d} \hat{f}(d\xi) \exp\left(-\frac{r(t-r)}{t} |\xi|^2\right) \left(\int_{\mathbb{R}^d} dw G_t(x-w) |\varphi(w)|\right)^2 \\
&\leq \int_0^t ds (t-s)^{\alpha-1} \int_0^s dr (s-r)^{-2\alpha} \int_{\mathbb{R}^d} \hat{f}(d\xi) \exp\left(-\frac{r(t-r)}{t} |\xi|^2\right) \int_{\mathbb{R}^d} dw G_t(x-w) |\varphi(w)|^2 \quad (84)
\end{aligned}$$

In this situation we need to be careful with $a(r)$. We know $a(r)$ is continuous on $(0, s]$ but also we know that $a(r) \rightarrow |\varphi(x)|^2$ as $r \rightarrow 0$ and thus this shows that $a(r)$ is bounded by some constant M_s on $[0, s]$.

Now we can see that (11.36) is bounded by

$$\begin{aligned}
(11.36) &\leq M_t \int_0^t ds (t-s)^{\alpha-1} \int_0^s dr (s-r)^{-2\alpha} \int_{\mathbb{R}^d} \hat{f}(d\xi) \exp\left(-\frac{r(t-r)}{t} |\xi|^2\right) \\
&= M_t \int_0^t dr \int_{\mathbb{R}^d} \hat{f}(d\xi) \exp\left(-\frac{r(t-r)}{t} |\xi|^2\right) \underbrace{\int_r^t ds (t-s)^{(\alpha-1)} (s-r)^{-2\alpha}}_{=K \cdot (t-r)^{-\alpha}} \quad (85)
\end{aligned}$$

$$= K \cdot M_t \int_0^t dr (t-r)^{-\alpha} \int_{\mathbb{R}^d} \hat{f}(d\xi) \exp\left(-\frac{r(t-r)}{t} |\xi|^2\right) \quad (86)$$

Nick's Question 11.7 Is (11.39) finite?

Le's Question 11.4 Here are some comments to your previous question. Firstly, did you forget a minus sign in exponential? This makes a big difference. Secondly, when $\alpha = 0$, this is finite, which is equivalent to Dalang's condition. But when $\alpha \in (0, 1)$, then finiteness of this quantity will impose a slightly stronger condition than Dalang's condition. Lastly, please use `\label{}` and `\eqref{}` to refer equations.

Yes I did mean for there to be a minus sign in all of the exponentials, I just fixed it. Also we will need $\alpha \in \left(0, \frac{1}{2}\right)$ in order for the underlined integral in (11.37) to be finite. α is also assumed to be less than $\frac{1}{2}$ in the Tessitore paper. Ill go back and use label and eqref.

Le's Question 11.5 It would be nice that α can be any value between 0 and $2 \wedge d$ (open interval), certainly if only that is the truth.

Nick's Question 11.8 Here are the details concerning $f(x) = \frac{1}{|x|^\beta}$. We want to ensure that

$$\int_0^s (s-r)^{-2\alpha} \int_{\mathbb{R}^d} \hat{f}(d\xi) \exp\left(-\frac{r(s-r)}{s} |\xi|^2\right) < \infty \quad (87)$$

And in this case

$$\hat{f}(x) = K \frac{1}{|x|^{d-\beta}}$$

and therefore (11.40) becomes

$$\int_0^s (s-r)^{-2\alpha} \int_{\mathbb{R}^d} d\xi \exp\left(-\frac{r(s-r)}{s} |\xi|^2\right) \frac{1}{|\xi|^{d-\beta}} \quad (88)$$

Now we apply a change of variables to (11.41). We let

$$\sqrt{\frac{r(s-r)}{s}} \xi_i = z_i \quad \implies \quad d\xi = \left(\frac{r(s-r)}{s}\right)^{-d/2} dz$$

and applying this to (11.41) transforms the integral into

$$\int_0^s (s-r)^{-2\alpha} \int_{\mathbb{R}^d} dz \frac{\exp(-|z|^2)}{|z|^2} s^{\beta/2} r^{-\beta/2} (s-r)^{-\beta/2} \quad (89)$$

L.C.: $|z|^{-2}$ should be $|z|^{\beta/2}$.

and using (11.40) this simplifies us to only show that

$$\int_0^s (s-r)^{-(2\alpha+\beta/2)} r^{-\beta/2} dr < \infty \quad (90)$$

$$\implies (2\alpha + \beta/2) < 1 \quad \text{and} \quad \beta/2 < 1 \quad (91)$$

However this seems to be an issue because we originally assumed that $\alpha \in (0, \frac{1}{2})$ and $\beta \in (0, 2 \wedge d)$ which would imply that

$$(2\alpha + \beta/2) \in (0, 2)$$

L.C.: here we are not going to prove that all Riesz kernels with $\beta \in (0, d \wedge 2)$ work. We need only to find one such Riesz kernel. Since $\alpha < 1/2$, so there any β such that

$$\beta < \min(2(1 - 2\alpha), d)$$

will do the job. The above argument is given α first and find a β . One can also argue reversely. Given a $\beta \in (0, 2 \wedge d)$, then we need to require

$$\alpha < \frac{1}{2}(1 - \beta/2).$$

Considering this issue, is it really necessary to have (11.40) be finite for every $\alpha \in (0, 1/2)$? Say we are given some Riesz Kernel $|x|^{-\beta_0}$ with $\beta_0 \in (0, d \wedge 2)$ then we can choose $\alpha \in (0, 1/2)$ small enough so that $(2\alpha + \beta_0/2) < 1$. One problem with that however is that in the Tessitore paper, they say that $\alpha \in (q^{-1}, 2^{-1})$. I need to see if this also needs to apply to our work or if we could say that $\alpha \in (0, 2^{-1})$. I believe we actually do need $\alpha \in (q^{-1}, 2^{-1})$ for the F_α operators to be compact. And because of this, then the biggest β_0 we can choose for the Riesz Kernel is such that $(q^{-1} + \beta_0/2) < 1$

Another thing to consider is then can we just let q be really big so given any $\beta_0 \in (0, d \wedge 2)$ then we just choose q to be large enough so that there exists an $\alpha \in (q^{-1}, 2^{-1})$ such that $(2\alpha + \beta_0/2) < 1$. I think this can be possible since as long as we can find some $\alpha \in (q^{-1}, 2^{-1})$ then we just continue along with that α in the proof and everything works since essentially α is a fixed value all throughout the proof. $q > 2$ is also just some fixed number throughout the proof so this also seems possible to just let q be big enough.

Nick's Question 11.9 In order for our solution to exist and to be in L_ρ^2 then we need to make several assumptions. (see page 13 of this document)

1. Dalang's condition needs to be met, hence we need $\Upsilon(\beta) < \infty$ for every $\beta > 0$.
(for one and hence for all $\beta > 0$).
2. The initial condition must satisfy $\int_{\mathbb{R}^d} e^{-a|x|^2} |\varphi(x)| dx < \infty$
Let's write $\int_{\mathbb{R}^d} e^{-a|x|^2} |\varphi|(dx) < \infty$ in case φ might be a measure.
3. Θ and $\Upsilon(0)$ from lemma 2.5 of [15] need to be finite and $\gamma_2 < 2\Upsilon(0)/\nu$

For the following we consider $f(x) = |x|^{-\beta}$ which means $\hat{f}(x) = C_{d,\beta} |x|^{\beta-d}$

Item 1: As long as $\beta \in (0, 2 \wedge d)$ then (1) is satisfied for $\beta > 0$

Item 2: Since we assume that $\varphi \in L_\rho^2$ then this implies that $\varphi \in L_\rho^1$ which further implies that $\varphi \in L_{\rho^*}^1$ where $\rho^* = e^{-|x|^2}$.

Item 3: This condition ensures that our solution is in the L_ρ^2 space where as (1) and (2) just ensure the existence and uniqueness of the solution. I believe that there is a problem with showing $\Upsilon(0) < \infty$. Considering $f(x) = |x|^{-\beta}$ then what we need is that

$$\int_{\mathbb{R}^d} \frac{f(x)}{|x|^{d-2}} dx = \int_{\mathbb{R}^d} \frac{\hat{f}(x)}{|x|^2} dx = \int_{\mathbb{R}^d} \frac{|x|^{\beta-d}}{|x|^2} dx = \int_{\mathbb{R}^d} \frac{1}{|x|^{d-\beta+2}} dx < \infty \quad (92)$$

However, the last integral in (11.45) is never finite.

L.C.: You are right. Please try the Bessel kernel as in Le's Question 8.1.

12 Discussions on Feb. 21–23

Le's Question 12.1 Here are several questions related to the weight functions:

1. Show that $\rho(\xi) = \exp(-r|\xi|)$ is admissible, i.e., Eq. (2.1) of [39] is satisfied. The constant $C_\rho(T) = e^{r^2 T/2}$ (for one dimensional case); check Proposition A.11 of [13].
2. Show that $\rho(\xi) = (1 + |\xi|^r)^{-1}$ is admissible in the sense of Eq. (2.1) of [39]. Moreover,

can you show that $C_\rho(T)$ is bounded function in T ?

3. Conditions for initial data seem to be: any nonnegative measures μ such that

$$\sup_{x \in \mathbb{R}^d} (\rho * \mu)(x) < +\infty.$$

13 Discussions on March 4

Nick's Question 13.1 As stated in the email, The inequality (11.27), and the justification of Fubini's theorem for the factorization theorem all assumed that

1. $|b(u)| < K|u|$
2. $\|u(r, z)\|_q \leq KJ_0(r, z)$

However in general for Lipschitz functions we can only say that $|b(u)| \leq K(1 + |u|)$ unless $b(0) = 0$. And for the moment bounds we have that $\|u(r, z)\|_q \leq K(1 + J_0(r, z))$ and it is not necessarily true that $K(1 + J_0(r, z)) \leq KJ_0(r, z)$. For example if $J_0(r, z) \rightarrow 0$ as $|z| \rightarrow \infty$ then there would not exist a K so $K(1 + J_0(r, z)) \leq KJ_0(r, z)$. But if $J_0(r, z)$ was bounded below by a number $k > 0$ then we could say that $K(1 + J_0(r, z)) \leq KJ_0(r, z)$ but I am not sure if this is always the case?

L.C.: You are right. In many cases, we have additional condition that sets $b(0) = 0$. This condition is required in order to keep the solution to be nonnegative (if the initial data is nonnegative).

So I tried to reprove the inequality in (11.27) using $|b(u)| \leq K(1 + |u|)$ and $\|u(r, z)\|_q \leq K(1 + J_0(r, z))$ and I came up with the following three things we need to assume to be finite which are:

1. $\int_0^s (s-r)^{-2\alpha} \int_{\mathbb{R}^d} \hat{f}(d\xi) \exp\left(-\frac{r(s-r)}{s}|\xi|^2\right) < \infty$
2. $\int_0^s (s-r)^{-2\alpha} \int_{\mathbb{R}^d} \hat{f}(d\xi) \exp(-(s-r)|\xi|^2) < \infty$
3. $\int_0^s (s-r)^{-2\alpha} \int_{\mathbb{R}^d} \hat{f}(d\xi) \exp\left(-\left(\frac{s-r}{2} + \frac{r(s-r)}{2s}\right)|\xi|^2\right) < \infty$

We already assumed that (1) was going to be finite but is that going to imply that (2) and (3) are going to be finite?

L.C.: Yes, most probably it is the case. We still need to prove that.

I have not yet tried to prove the justification of Fubini's Theorem yet with the assumptions that $|b(u)| \leq K(1 + |u|)$ and $\|u(r, z)\|_q \leq K(1 + J_0(r, z))$ but I am pretty sure that I will once again come up with those three things above that need to be finite.

I just went through all the details for justifying Fubini's theorem and again came up with needing (1), (2) and (3) above to be finite. I have a lot of pages of calculations handwritten in my notebook and will type them all up

References

- [1] Amir, Gideon, Ivan Corwin and Jeremy Quastel. Probability distribution of the free energy of the continuum directed random polymer in $1+1$ dimensions. *Comm. Pure Appl. Math.* 64 (2011), no. 4, 466–537.

- [2] Assing, Sigurd and Ralf Manthey. Invariant measures for stochastic heat equations with unbounded coefficients. *Stochastic Process. Appl.* 103 (2003), no. 2, 237–256.
- [3] Balan, Raluca M. and Le Chen. Parabolic Anderson model with space-time homogeneous Gaussian noise and rough initial condition *J. Theoret. Probab.*, to appear 2017.
- [4] V. Bally and E. Pardoux. Malliavin calculus for white noise driven parabolic SPDEs. *Potential Anal.* 9 (1998), no. 1, 27–64.
- [5] Bertini, Lorenzo and Giambattista Giacomin. Stochastic Burgers and KPZ equations from particle systems. *Comm. Math. Phys.* 183 (1997), no. 3, 571–607.
- [6] Bertini, Lorenzo and Giambattista Giacomin. On the long-time behavior of the stochastic heat equation. *Probab. Theory Related Fields*, 114 (1999), no. 3, 279–289.
- [7] Bouleau, Nicolas and Francis Hirsch. *Dirichlet forms and analysis on Wiener space*. De Gruyter Studies in Mathematics, 14. Walter de Gruyter & Co., Berlin, 1991. x+325 pp.
- [8] Cannizzaro, Giuseppe, Peter K. Friz and Paul Gassiat. Malliavin Calculus for regularity structures: the case of gPAM. *J. Funct. Anal.*, 272 (2017), no. 1, 363–419.
- [9] Carmona, René A. and Stanislav A. Molchanov. Parabolic Anderson problem and intermittency. *Mem. Amer. Math. Soc.*, 108 (1994), no. 518, viii+125 pp.
- [10] Cerrai, Sandra. *Second order PDE's in finite and infinite dimension*. A probabilistic approach. Lecture Notes in Mathematics, 1762. Springer-Verlag, Berlin, 2001. x+330 pp.
- [11] Cerrai, Sandra. Stochastic reaction-diffusion systems with multiplicative noise and non-Lipschitz reaction term. *Probab. Theory Related Fields* 125 (2003), no. 2, 271–304.
- [12] Cerrai, Sandra. Asymptotic behavior of systems of stochastic partial differential equations with multiplicative noise. *Stochastic partial differential equations and applications–VII*, 61–75, Lect. Notes Pure Appl. Math., 245, Chapman & Hall/CRC, Boca Raton, FL, 2006.
- [13] Chen, Le and Robert C. Dalang. Moments and growth indices for nonlinear stochastic heat equation with rough initial conditions. *Ann. Probab.* 43 (2015), no. 6, 3006–3051.
- [14] Chen, Le, Yaozhong Hu and David Nualart. Regularity and strict positivity of densities for the nonlinear stochastic heat equation. *Mem. Amer. Math. Soc.*, 2019, to appear.
- [15] Chen, Le and Jingyu Huang. Comparison principle for stochastic heat equation on $\mathbb{R}^d$. *Ann. Probab.* 2019, to appear.
- [16] Chen, Le and Kunwoo Kim. Nonlinear stochastic heat equation driven by spatially colored noise: moments and intermittency. *Acta Math. Sci. Ser. B*, 2019, to appear.
- [17] Dalang, Robert C. Extending the martingale measure stochastic integral with applications to spatially homogeneous s.p.d.e.'s. *Electron. J. Probab.* 4 (1999), no. 6, 29 pp.
- [18] Dalang, Robert, Davar Khoshnevisan, Carl Mueller, David Nualart, and Yimin Xiao *A minicourse on stochastic partial differential equations*. Held at the University of Utah, Salt Lake City, UT, May 8–19, 2006. Edited by Khoshnevisan and Firas Rassoul-Agha. Lecture Notes in Mathematics, 1962. Springer-Verlag, Berlin, 2009. xii+216 pp.
- [19] Dalang, Robert C. and Lluís Quer-Sardanyons. Stochastic integrals for spde's: a comparison. *Expo. Math.* 29 (2011), no. 1, 67–109.
- [20] Da Prato, Giuseppe and Jerzy Zabczyk. *Stochastic equations in infinite dimensions*. Encyclopedia of Mathematics and its Applications, 44. Cambridge University Press, Cambridge, 1992. xviii+454 pp.
- [21] Da Prato, Giuseppe and Jerzy Zabczyk. *Ergodicity for infinite-dimensional systems*. London Mathematical Society Lecture Note Series, 229. Cambridge University Press, Cambridge, 1996.

- [22] Eckmann, Jean-Pierre and Martin Hairer. Invariant measures for stochastic partial differential equations in unbounded domains. *Nonlinearity* 14 (2001), no. 1, 133–151.
- [23] Foondun, Mohammud and Davar Khoshnevisan. On the stochastic heat equation with spatially-colored random forcing. *Trans. Amer. Math. Soc.* 365 (2013), no. 1, 409–458.
- [24] Grafakos, Loukas *Modern Fourier analysis*. Third edition. Graduate Texts in Mathematics, 250. Springer, New York, 2014. xvi+624 pp.
- [25] Goldys, Benjamin, and Bohdan Maslowski. Uniform exponential ergodicity of stochastic dissipative systems. *Czechoslovak Math. J.* 51(126) (2001), no. 4, 745–762.
- [26] Hairer, Martin. A theory of regularity structures. *Invent. Math.*, 198 (2) (2014), 1–236.
- [27] Hairer, Martin. Solving the KPZ equation. *Ann. of Math.*, (2) 178 (2013), no. 2, 559–664.
- [28] Y. Hu, J. Huang, D. Nualart, X. Sun. Smoothness of the joint density for spatially homogeneous SPDEs. *J. Math. Soc. Japan*, Vol. 67, No. 4 (2015), 1605–1630.
- [29] Maslowski, Bohdan and Jan Seidler. On sequentially weakly Feller solutions to SPDE's. *Atti Accad. Naz. Lincei Cl. Sci. Fis. Mat. Natur. Rend. Lincei (9) Mat. Appl.* 10 (1999), no. 2, 69–78.
- [30] Misiats, Oleksandr, Oleksandr Stanzhytskyi, and Nung-Kwan Yip. Existence and uniqueness of invariant measures for stochastic reaction-diffusion equations in unbounded domains. *J. Theoret. Probab.* 29 (2016), no. 3, 996–1026.
- [31] Mueller, Carl and David Nualart. Regularity of the density for the stochastic heat equation. *Electron. J. Probab.*, 13 (2008), no. 74, 2248–2258.
- [32] D. Nualart. *Malliavin calculus and its applications*. CBMS Regional Conference Series in Mathematics, 110. Published for the Conference Board of the Mathematical Sciences, Washington, DC; by the American Mathematical Society, Providence, RI, 2009.
- [33] D. Nualart and L. Quer-Sardanyons. Existence and smoothness of the density for spatially homogeneous SPDEs. *Potential Anal.* 27 (2007), no. 3, 281–299.
- [34] E. Nualart. On the density of systems of non-linear spatially homogeneous SPDEs. *Stochastics*, 85 (2013), no. 1, 48–70.
- [35] F. W. J. Olver, D. W. Lozier, R. F. Boisvert, and C. W. Clark, editors. *NIST handbook of mathematical functions*. U.S. Department of Commerce National Institute of Standards and Technology, Washington, DC. Cambridge Univ. Press, Cambridge, 2010.
- [36] Pardoux, Etienne and Tusheng Zhang. Absolute continuity of the law of the solution of a parabolic SPDE. *J. Funct. Anal.* 112 (1993), no. 2, 447–458.
- [37] Peszat, and Zabczyk Stochastic Evolution Equations with a Spatially Homogenous Wiener Processes. *SNS* (1995)
- [38] Quer-Sardanyons, Lluís and Marta Sanz-Solé. A stochastic wave equation in dimension 3: smoothness of the law. *Bernoulli*, 10 (2004), no. 1, 165–186.
- [39] Tessitore, Gianmario and Jerzy Zabczyk. Invariant measures for stochastic heat equations. *Probab. Math. Statist.* 18 (1998), no. 2, Acta Univ. Wratislav. No. 2111, 271–287.
- [40] Walsh, John B. *An Introduction to Stochastic Partial Differential Equations*. In: *École d'été de probabilités de Saint-Flour, XIV—1984*, 265–439. Lecture Notes in Math. 1180, Springer, Berlin, 1986.